

⚡ Quick Reference Guide



Start in 3 Commands

```
cd /home/sujay/Programming/IDS
chmod +x run_afpacket_mode.sh
sudo ./run_afpacket_mode.sh
```

Select option 1 → Pipeline starts! ✓



Command Cheat Sheet

Basic Operations

Task	Command
Start Pipeline	sudo ./run_afpacket_mode.sh start
Check Status	sudo ./run_afpacket_mode.sh status
View Logs	sudo ./run_afpacket_mode.sh logs
Stop Pipeline	sudo ./run_afpacket_mode.sh stop
Interactive Menu	sudo ./run_afpacket_mode.sh

Component Specific

Task	Command
Start Only Kafka	sudo ./run_afpacket_mode.sh kafka
Start Only Suricata	sudo ./run_afpacket_mode.sh suricata
Start Only ML Consumer	sudo ./run_afpacket_mode.sh ml
Start Only Kafka Bridge	sudo ./run_afpacket_mode.sh bridge

DPDK Mode

Task	Command
Start DPDK Pipeline	sudo ./run_dpdk_mode.sh start
Bind Interface	sudo ./run_dpdk_mode.sh bind
Unbind Interface	sudo ./run_dpdk_mode.sh unbind
Check DPDK Status	sudo ./run_dpdk_mode.sh info

🔍 Verify Services

Check if Services are Running

```
# Kafka
ps aux | grep kafka | grep -v grep

# Suricata
ps aux | grep suricata | grep -v grep

# ML Consumer
ps aux | grep ml_kafka_consumer | grep -v grep

# Kafka Bridge
ps aux | grep suricata_kafka_bridge | grep -v grep
```

Quick Status Check

```
# AF_PACKET mode
sudo ./run_afpacket_mode.sh status

# DPDK mode
sudo ./run_dpdk_mode.sh status
```

📊 View Output Data

Suricata Alerts (Real-time)

```
tail -f /var/log/suricata/eve.json
```

ML Predictions (Real-time)

```
python3 << 'EOF'
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'ml-predictions',
    bootstrap_servers=['localhost:9092'],
    auto_offset_reset='latest',
    value_deserializer=lambda x: json.loads(x.decode('utf-8'))
)
```

```
for message in consumer:
    print(json.dumps(message.value, indent=2))
EOF
```

Kafka Topics

```
# List all topics
kafka-topics.sh --bootstrap-server localhost:9092 --list

# Read messages from suricata-alerts
kafka-console-consumer.sh --bootstrap-server localhost:9092 \
    --topic suricata-alerts --from-beginning

# Read messages from ml-predictions
kafka-console-consumer.sh --bootstrap-server localhost:9092 \
    --topic ml-predictions --from-beginning
```

Testing

Generate Test Traffic

```
cd /home/sujay/Programming/IDS/tests

# Benign traffic
python3 test_benign_traffic.py

# Attack traffic
python3 test_attack_generator.py

# ML testing
python3 test_ml_classifications.py

# DPDK testing
python3 test_dpdk_scapy_integration.py
```

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⚙️ Configuration

Edit Configuration

```
nano
/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/config/pipeline.conf
```

Key Settings

Setting	Location	Default	Purpose
Network Interface	NETWORK_INTERFACE	enx00e04c36074c	Packet capture interface
Suricata Threads	SURICATA_CORES	2	Detection threads
ML Batch Size	ML_BATCH_SIZE	100	Batch inference size
ML Threshold	ML_CONFIDENCE_THRESHOLD	0.7	Alert threshold
Kafka Servers	KAFKA_BOOTSTRAP_SERVERS	localhost:9092	Kafka location

Common Issues

Issue: "Permission denied"

```
sudo chmod +x /home/sujay/Programming/IDS/run_afpacket_mode.sh
```

Issue: "Interface not found"

```
ip link show # Find your interface
# Edit: dpdk_suricata_ml_pipeline/config/pipeline.conf
```

Issue: "Kafka already in use"

```
sudo lsof -i :9092 # Find process
sudo kill -9 <PID> # Kill it
```

Issue: "No alerts generated"

```
# Check Suricata logs
tail -f /var/log/suricata/suricata.log

# Check if traffic is captured
sudo tcpdump -i YOUR_INTERFACE -c 10

# Check Kafka topics
kafka-topics.sh --bootstrap-server localhost:9092 --list
```

Issue: High CPU usage

```
# Increase threads
nano dpdk_suricata_ml_pipeline/config/pipeline.conf
# Set: SURICATA_CORES="4" (or higher)

# Restart:
sudo ./run_afpacket_mode.sh stop
sudo ./run_afpacket_mode.sh start
```

 Documentation Files

File	Purpose
HOW_TO_RUN.md	Step-by-step setup guide ← You are here
PIPELINE_ARCHITECTURE.md	Technical architecture details
NEXT_STEPS.md	Project roadmap and future plans
FINAL_SUMMARY.md	Cleanup report
README.md	Main documentation
QUICKSTART.md	Quick setup guide
USB_ADAPTER_GUIDE.md	USB adapter specific setup
PRODUCTION_DPDK_GUIDE.md	DPDK production deployment

 Pro Tips

1. Keep Multiple Terminals

```
# Terminal 1: Monitor status
watch -n 1 'sudo ./run_afpacket_mode.sh status'

# Terminal 2: View logs
tail -f /var/log/suricata/eve.json

# Terminal 3: Generate test traffic
python3 tests/test_benign_traffic.py

# Terminal 4: Run commands
# available for other tasks
```

2. Background Execution

```
# Run pipeline in background
nohup sudo ./run_afpacket_mode.sh start > pipeline.log 2>&1 &

# Check log
tail -f pipeline.log

# Stop later
sudo ./run_afpacket_mode.sh stop
```

3. Autostart on Boot (Linux)

```
# Create systemd service
sudo cat > /etc/systemd/system/ids-pipeline.service << EOF
[Unit]
Description=IDS Pipeline
After=network.target

[Service]
Type=simple
WorkingDirectory=/home/sujay/Programming/IDS
ExecStart=/home/sujay/Programming/IDS/run_afpacket_mode.sh start
Restart=always

[Install]
WantedBy=multi-user.target
EOF

# Enable and start
sudo systemctl enable ids-pipeline.service
sudo systemctl start ids-pipeline.service

# Check status
sudo systemctl status ids-pipeline.service
```

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4. Docker Deployment (Optional)

```
# If containerization desired, add Dockerfile to project
docker build -t ids-pipeline .
docker run -d --net=host ids-pipeline run_afpacket_mode.sh start
```

Quick Scenarios

Scenario 1: Basic Testing

```
# Terminal 1
cd /home/sujay/Programming/IDS
sudo ./run_afpacket_mode.sh start

# Terminal 2 (after 10 seconds)
cd /home/sujay/Programming/IDS/tests
python3 test_benign_traffic.py

# Terminal 1
sudo ./run_afpacket_mode.sh logs
# Should see alerts and ML predictions
```

Scenario 2: Performance Benchmarking

```
# Start pipeline
sudo ./run_afpacket_mode.sh start

# Monitor performance
watch -n 1 'ps aux | grep suricata | head -1'

# Generate high-volume traffic
cd tests/
for i in {1..10}; do
    python3 test_attack_generator.py &
done
wait

# Check performance
top -p $(pgrep suricata)
```

Scenario 3: DPDK Mode Testing

```
# If DPDK-compatible hardware available
sudo ./run_dpdk_mode.sh start

# Monitor interface binding
sudo ./run_dpdk_mode.sh info

# Generate traffic
cd tests/
python3 test_dpdk_scapy_integration.py

# Check results
tail -f /var/log/suricata/eve.json
```

For detailed setup help:

- Read: [HOW_TO_RUN.md](#) (you are here)
- Architecture: [PIPELINE_ARCHITECTURE.md](#)
- Quick start: [QUICKSTART.md](#)

For architecture questions:

- Read: [PIPELINE_ARCHITECTURE.md](#)
- Check: [FLOW_BASED_ML_ARCHITECTURE.md](#)

For future improvements:

- See: [NEXT_STEPS.md](#)

For troubleshooting:

- Check issues in: [HOW_TO_RUN.md](#) (Troubleshooting section)
- View logs: `sudo ./run_afpacket_mode.sh logs`

 Success Checklist

- ☐ Installed dependencies
- ☐ Configured network interface
- ☐ Started pipeline
- ☐ All services running
- ☐ Generated test traffic
- ☐ Received alerts
- ☐ ML predictions working
- ☐ Understand architecture (read [PIPELINE_ARCHITECTURE.md](#))
- ☐ Reviewed next steps (read [NEXT_STEPS.md](#))

You're all set! Start with: `sudo ./run_afpacket_mode.sh start` 